



Alice Rogier

Postdoctoral researcher
in Medical Informatics

Contact

29 years old
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Education

2020–2024 PhD in Medical Informatics *Université Paris Cité*

Biomedical data science, ontologies, knowledge extraction from EHRs, semantic web, temporal reasoning, NLP, graph mining, survival analysis.

2017–2019 MSc in Bioinformatics *Université Paris-Saclay*

Algorithmics, statistics (frequentist and Bayesian), machine learning, data mining, database managing, omics data analysis (genomics, transcriptomics, structural biology), programming (Python, Java, C, R, Perl)

2016–2017 BSc in Organism Biology and Ecology *Université Paris-Saclay*

Ecology and evolution, biodiversity, molecular and cellular biology, environmental data analysis

Professional experiences

Since Postdoctoral researcher *ITiSB, Viña del mar, Universidad Andres Bello*

Sep 2025 I am currently a postdoctoral researcher initiating a collaboration with the ITiSB. I arrived in Chile in September, which allowed me to gain a deeper understanding of the local research community and opportunities, as well as to integrate into the UNAB ecosystem.

Nov 2024 Postdoctoral researcher *Carpem Cancer Institute, Paris*

Jul 2025 I was a post-doctoral researcher at the *Carpem Cancer Institute* in Paris, focusing on structuring and integrating chemotherapy data into a database compatible with the *Master Observational Trial* methodology. This approach aims to facilitate observational studies by combining data from both clinical trials and high-quality electronic health records to support robust and meaningful analyses. A work is in preparation on these results.

Dec 2020 PhD student *Inserm-Inria HeKA Team, Paris*

Oct 2024 I completed my PhD in computer science working within the HeKA team, an interdisciplinary laboratory of approx. 20 professors, which combines expertise from mathematicians, clinicians, statisticians, and computer scientists to address challenges in biomedical informatics. I was also associated with Paris's Hospitals (AP-HP). During my PhD, I specialized in extracting and predicting chemotherapy responses from Electronic Health Records (EHRs). My key contributions include:

- *OntoTox*: An ontology organizing and integrating knowledge about chemotherapy toxicities, extracted from diverse sources in EHRs (medical notes, tables and questionnaires) [4].
- *ChemoOnto*: An ontology representing both patient chemotherapy courses and theoretical protocols, enabling standardization and cross-hospital comparisons [2].
- *ProtoDrift*: A metric to quantify deviations in chemotherapy protocols, validated through survival analysis in two hospitals, supporting precision oncology research [1].

Professional experiences (continued)

Nov 2019 **Design engineer** *Inserm, Paris Georges Pompidou hospital*

Dec 2020 I took part in the QualiHealth project aimed at improving the quality of healthcare data. I developed NLP tools and an API annotation pipeline for classifying COVID-19 patient phenotypes from medical narrative reports [6]. In parallel, I collaborated with general practitioners to implement OntoDol [5], a rule-based ontology for chronic pain triage mapped to SNOMED-CT. This experience highlighted my ability to work with researchers and clinicians from diverse backgrounds.

Mar - Aug 2019 **Intern** *Inserm, Paris Georges Pompidou hospital*

I worked on high-throughput phenotyping for glomerular diseases through clinical notes.

Jul - Aug 2018 **Intern** *Parasitology Unit of the Center Dr. Hideyo, Mexico*

I worked on a project aimed at identifying Chagas disease blood meals from *Triatomia dimidiata* in the Yucatán, using PCR and electrophoresis.

Jun - Aug 2017 **Intern** *EcoSys laboratory, INRA.*

Characterization of metal contaminations in urban farm soils.

Scientific publications

- [1] **Alice Rogier**, E. Ashton, C. Barat, R. Griffier, V. Jouhet, E. Paillaud, B. Sabatier, J. Taïeb, É. Zapletal, B. Rance, and A. Coulet, “ProtoDrift: Evaluating the Impact of Small-to-Big Adjustments to Chemotherapy Protocols,” 2025, Preprint available at medRxiv:2025.05.09.25327311.
- [2] J. H. Jhee, **Alice Rogier**, D. Giraud, E. Pinet, B. Sabatier, B. Rance, and A. Coulet, “Representation and comparison of chemotherapy protocols with ChemoKG and graph embeddings,” in *WAT4HCLS 2024 – 15th International Conference on Semantic Web Applications and Tools for Health Care and Life Sciences*, vol. 3890, 2024, pp. 23–35.
- [3] A. Neuraz, G. Vaillant, C. Arias, O. Birot, K.-T. Huynh, T. Fabacher, **Alice Rogier**, N. Garcelon, I. Lerner, B. Rance, et al., “Facilitating phenotyping from clinical texts: the medkit library,” *Bioinformatics*, vol. 40, no. 12, btae681, Dec. 2024.
- [4] **Alice Rogier**, A. Coulet, and B. Rance, “Using an ontological representation of chemotherapy toxicities for guiding the information extraction and integration from EHRs,” in *MEDINFO 2021: One World, One Health–Global Partnership for Digital Innovation*, vol. 290, Studies in Health Technology and Informatics, IOS Press, 2022, pp. 91–95.
- [5] A. Saadi, **Alice Rogier**, A. Burgun, and R. Tsopra, “Design of an Ontology-Based Triage System for Patients with Chronic Pain,” in *MEDINFO 2021: One World, One Health–Global Partnership for Digital Innovation*, vol. 290, Studies in Health Technology and Informatics, IOS Press, 2022, pp. 81–85.
- [6] A. Neuraz, I. Lerner, W. Digan, N. Paris, R. Tsopra, **Alice Rogier**, D. Baudoin, K. B. Cohen, A. Burgun, N. Garcelon, et al., “Natural language processing for rapid response to emergent diseases: case study of calcium channel blockers and hypertension in the COVID-19 pandemic,” *Journal of medical Internet research*, vol. 22, no. 8, e20773, Aug. 2020.

Programming languages

R, Python

Query languages

SQL, SPARQL

Human languages

French (native speaker)

Fluent in English and Spanish

Research expertise

Knowledge graph, Survival analysis, Statistical learning, Text mining

Hobbies

Modern jazz

Hiking

Talk and events

I presented my work in contributed talks to two international conferences (the 2023 International Conference on Intelligent Systems for Molecular Biology and MedInfo 2021), where I was awarded the second prize for best student paper. I also presented my work in other conferences and workshops at the local level. Committed to open science, I make all my slides and posters available on my personal website.

Type	Event	Date
Poster	AI4Health 2021	04/01/2021
Oral presentation	HeKA team meeting	15/02/2021
Oral presentation	HeKA team meeting	13/09/2021
Poster	CRC scientific day	23/09/2021
Oral presentation	MedInfo 2021	02/10/2021
Poster	AI4Health 2022	12/01/2022
Oral presentation	Bernoulli Lab kick-off (APHP-Inria)	22/03/2022
Oral presentation	CRC scientific day	22/09/2022
Poster	ISMB/ECCB 2023 Lyon	25/07/2023
Oral presentation	ISMB/ECCB 2023 Lyon	25/07/2023

Teaching

Spring 2022, Database Management (L3) Université Paris Cité

24h I taught at the University of Paris-Cité in the Informatics department. In this course, I led practical work on database design and query languages for third-year computer science students.

Falls 2021 and Python Programming (L1) Université Paris Cité

2022, I led first-year MIAH (Applied Mathematics for Social Sciences) students through foundational
48h Python programming.

Spring 2023, GUI Projects in Java (L2) Université Paris Cité

24h I guided groups of 5–6 second-year computer science students through Java-based Graphical User Interface (GUI) projects I designed, AlignSeq and PhyloApp, introducing them to biological sequence alignment and phylogeny. Advanced students progressed to a fusion project combining both, in collaboration with another group.

Awards



Second Best Student Paper, MedInfo Conference 2021

References

Bastien Rance Assistant Professor in Medical Informatics, Georges Pompidou Hospital, Université Paris Cité
Researcher at the Must team (Inserm)
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Adrien Coulet Assistant Professor in Informatics at Université Paris Cité and Université de Lorraine
Researcher at the HeKA team (Inserm-Inria)
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